LANS: actual PPO training after the audit fixes

27 October2026. The user explicitly requested training after the failed causal-planner headroom diagnostic. This new finite protocol proceeds despite that heuristic failure; historical outcomes and criteria remain unchanged. The20% target is measured, not guaranteed.

What was actually trained

Three new RNG seeds continue the corresponding previous causal-history PPO checkpoints for exactly100,000 additional environment steps each:300,000 NEW steps total. Actor, critic, Adam optimizer and reward-normalization statistics are retained; source preservation is contract-tested and initial checkpoints/normalizers are saved. The rollout buffer/environment/RNG are fresh. Original supervised and previous PPO training budgets are additional. Only the three final endpoints are compared: no best-checkpoint cherry-picking, per-seed mixing or post-result extensions.

Training uses the corrected v10 terminal next-observation interface, strict packet/identity checks and nominal RK4 predictor with average actual actuator-rate cost. Source/model input dependencies are pinned to the prior immutable evidence. The causal20-channel input ABI, physical reward, submitted-request actor credit, fixed-step GAE, all-transition critic,50ms stale cutoff, residual bound0.08rad, filter0.3s, cap0.25, recovery/ramp/margin and actuator limits remain unchanged. No future response/service information is provided to the actor.

The training distribution covers15/20/25m/s, two known paths, exactmirrors and nominal/tire/20/40/100ms conditions. Its noise namespace is new and separate from the fresh development namespace. Severe-delay transitions can have zero policy authority; they are retained rather than removed to inflate tracking gains. No reserved path is opened.

Pair	New RNG seed	Additional steps	Submitted requests	Active assistance ticks	Wall seconds
0	7701	100000	55471	2978	68.0
1	7802	100000	59045	2709	68.7
2	7903	100000	52861	3996	63.7

Fresh matched evaluation

The predeclared full development grid has 840 episodes:15/25m/s x2knownpaths x2mirrors x5scenarios x7sources x3actor/noise groups. Every source uses the same corrected supervisor, sensing, actuator limits and matched exogenous conditions. Student and oldPPO weights remain frozen; the newPPO is the continued endpoint. The causal nominal planner and both fixed corrections are nonlearned controls. A full robust published vehicle baseline/MPC is not included, so superiority over all published controllers cannot be claimed.

Independent single-observation inference and full input/history/compensation/packet/recovery/physical/rewar replay verify 1,153,740 transitions and 281,880 neural decisions. Planner proposals are independently regenerated from available information; 144 severe-delay/LQR trajectory pairs match exactly. All input source/model hashes and exports are checked.

Source	Tracking gain vs LQR	Heading change	Steering-rate RMS change	Survival
lqr	+0.000%	+0.000%	+0.000%	120/120
negative	-0.591%	-1.247%	-1.075%	120/120
positive	-0.591%	-1.247%	-1.075%	120/120
student_rk	+0.429%	-0.167%	+0.143%	120/120

Source	Tracking gain vs LQR	Heading change	Steering-rate RMS change	Survival
oldppo_rk	+0.360%	-0.196%	+0.036%	120/120
planner_rk	+1.025%	-1.167%	-2.182%	120/120
newppo_rk	+0.098%	-0.223%	-0.074%	120/120

New PPO contribution and20% target

- NewPPO tracking gain vs lqr: **+0.0976%**.
- NewPPO tracking gain vs negative: **+0.6846%**.
- NewPPO tracking gain vs positive: **+0.6846%**.
- NewPPO tracking gain vs student_rk: **-0.3328%**.
- NewPPO tracking gain vs oldppo_rk: **-0.2630%**.
- NewPPO tracking gain vs planner_rk: **-0.9373%**.

20% development screen: **False**. Failed checks: less_than20pct_vs_lqr; nonpositive_pair2_vs_lqr; less_than20pct_vs_negative; less_than20pct_vs_positive; less_than20pct_vs_student_rk; nonpositive_pair0_vs_student_rk; nonpositive_pair1_vs_student_rk; nonpositive_pair2_vs_student_rk; less_than20pct_vs_oldppo_rk; nonpositive_pair0_vs_oldppo_rk; nonpositive_pair2_vs_oldppo_rk; less_than20pct_vs_planner_rk; nonpositive_pair0_vs_planner_rk; nonpositive_pair1_vs_planner_rk; nonpositive_pair2_vs_planner_rk.

The screen requires at least20% lower mean penalized episode tracking RMSE than EVERY listed comparator, positive paired gain in all three groups, no survival loss, nominal tracking no more than2% worse than LQR and overall heading/actuator-rate RMS no more than5% higher. Exact ratio comparisons avoid floating-point misclassification at the20% boundary. Aggregate metrics do not guarantee every individual speed/path cell improves.

The matched old/new PPO comparison measures the result of additional training under the corrected environment. It does not isolate the individual causal effects of terminal-observation repair, predictor integration, continued optimizer/reward statistics or RNG changes. Prior0.379%/1.488% diagnostic values came from a narrower grid with other noise; do not compare those aggregates as if only training changed.

Limited authority, sparse accepted proposals and a strong LQR/student/planner remain structural constraints. A failed target is not converted into a pass by deleting delayed conditions, weakening baselines, extending training until a score crosses a line or choosing different best seeds. No universal impossibility claim follows from this finite study.

Timing, confirmation and scope

Winner-specific original-deadline timing is conditional on the20% selection screen. If the screen fails, no current-model timing success is claimed. Previous v9 zero-miss timing and the v10 one-miss smoke are historical experiments, not certification of these new actors. All reserved/native Webots confirmation, actual laptop execution and physical sensor/communication/actuator validation remain separate evidence requirements. No deployment, global stability, universal novelty, energy or49-fold speedup claim is made.

Resume

```
python tools/reviewer_v11/run11.py --publish
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The command checks provenance/contracts, continues the three finite runs, exports actors, executes the fixed matched matrix, independently replays raw data, seals evidence and builds Word/PDF reports. Completed and partial outputs are preserved. Exit2 means the20% empirical requirement remains unmet. No further training starts automatically.